

```
andersondeaah — PC2 — telnet 172.16.53...

Trying 172.16.53.128...
Connected to 172.16.53.128.
[Escape character is '^']
show ip

NAME       : PC2[1]
IP/MASK     : 10.11.15.66/27
GATEWAY     : 10.11.15.65
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 20030
RHOST:PORT  : 127.0.0.1:20031
MTU         : 1500

PC2> 
```

```
andersondeaah — R1 — telnet 172.16.53.128 5000 — 80x24

changed state to up
[R1_00111522#
[R1_00111522#show ip
% Incomplete command.

[R1_00111522#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
C       10.11.15.0/26 is directly connected, FastEthernet0/0
C       10.11.15.64/27 is directly connected, FastEthernet0/1
C       10.11.15.96/28 is directly connected, FastEthernet2/0
S       10.11.15.124/30 [1/0] via 10.11.15.122
C       10.11.15.120/30 is directly connected, Serial1/0
S       10.11.15.112/29 [1/0] via 10.11.15.122
R1_00111522#
```

andersondeaah — PC1 — telnet 172.16.53.128 5003 — 80x24

Trying 172.16.53.128...
Connected to 172.16.53.128.
Escape character is '^']

[show ip

NAME : PC1[1]
IP/MASK : 10.11.15.2/26
GATEWAY : 10.11.15.1
DNS :
MAC : 00:50:79:66:68:00
LPORT : 20028
RHOST:PORT : 127.0.0.1:20029
MTU : 1500

PC1> █

andersondeaah — PC3 — telnet 172.16.53.128 5007 — 80x24

Trying 172.16.53.128...
Connected to 172.16.53.128.
Escape character is '^']

[show ip

NAME : PC3[1]
IP/MASK : 10.11.15.98/28
GATEWAY : 10.11.15.97
DNS :
MAC : 00:50:79:66:68:02
LPORT : 20032
RHOST:PORT : 127.0.0.1:20033
MTU : 1500

PC3> █

andersondeaah — PC4 — telnet 172.16.53.128 5009 — 80x24

```
Trying 172.16.53.128...
Connected to 172.16.53.128.
Escape character is '^]'.
[show ip
```

```
NAME       : PC4[1]
IP/MASK    : 10.11.15.114/29
GATEWAY    : 10.11.15.113
DNS        :
MAC        : 00:50:79:66:68:03
LPORT     : 20034
RHOST:PORT : 127.0.0.1:20035
MTU        : 1500
```

```
PC4> █
```

andersondeaah — R2 — telnet 172.16.53.128 5001 — 80x24

```
[R2_00111522#
[R2_00111522#show ip routes
      ^
% Invalid input detected at '^' marker.
```

```
[R2_00111522#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S    10.11.15.0/26 [1/0] via 10.11.15.121
S    10.11.15.64/27 [1/0] via 10.11.15.121
S    10.11.15.96/28 [1/0] via 10.11.15.121
C    10.11.15.124/30 is directly connected, Serial1/1
C    10.11.15.120/30 is directly connected, Serial1/0
S    10.11.15.112/29 [1/0] via 10.11.15.126
R2_00111522# █
```

andersondeaah — R3 — telnet 172.16.53.128 5002 — 80x24

```
R3_00111522#
*Mar 1 00:45:42.643: %SYS-5-CONFIG_I: Configured from console by console
R3_00111522#write memory
Building configuration...
[OK]
R3_00111522#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S    10.11.15.0/26 [1/0] via 10.11.15.125
S    10.11.15.64/27 [1/0] via 10.11.15.125
S    10.11.15.96/28 [1/0] via 10.11.15.125
C    10.11.15.124/30 is directly connected, Serial1/1
S    10.11.15.120/30 [1/0] via 10.11.15.125
C    10.11.15.112/29 is directly connected, FastEthernet0/0
R3_00111522#
```

andersondeaah — R1 — telnet 172.16.53.128 5000 — 80x24

```
C    10.11.15.0/26 is directly connected, FastEthernet0/0
C    10.11.15.64/27 is directly connected, FastEthernet0/1
C    10.11.15.96/28 is directly connected, FastEthernet2/0
S    10.11.15.124/30 [1/0] via 10.11.15.122
C    10.11.15.120/30 is directly connected, Serial1/0
S    10.11.15.112/29 [1/0] via 10.11.15.122
R1_00111522#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Prot
FastEthernet0/0	10.11.15.1	YES	manual	up	up
FastEthernet0/1	10.11.15.65	YES	manual	up	up
Serial1/0	10.11.15.121	YES	manual	up	up
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down
FastEthernet2/0	10.11.15.97	YES	manual	up	up

```
R1_00111522#
```

```
andersondeaah — R2 — telnet 172.16.53.128 5001 — 80x24

10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S    10.11.15.0/26 [1/0] via 10.11.15.121
S    10.11.15.64/27 [1/0] via 10.11.15.121
S    10.11.15.96/28 [1/0] via 10.11.15.121
C    10.11.15.124/30 is directly connected, Serial1/1
C    10.11.15.120/30 is directly connected, Serial1/0
S    10.11.15.112/29 [1/0] via 10.11.15.126
R2_00111522#show ip interface brief
Interface          IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0    unassigned      YES unset    administratively down down
FastEthernet0/1    unassigned      YES unset    administratively down down
Serial1/0          10.11.15.122    YES manual    up
Serial1/1          10.11.15.125    YES manual    up
Serial1/2          unassigned      YES unset    administratively down down
Serial1/3          unassigned      YES unset    administratively down down
R2_00111522#
```


Ayudando a construir hogares

00 11 15 22

1. Carinet ↑

A: 11

B: 15

Red base asignado: 10.11.15.0/24

$$2^n - 2 \geq \text{Hosts}$$

$$\text{mask} = 32 - n$$

$$\text{salto} = 2^n$$

50

25

12

6

2

2

Ej. 1:

Red	Red Asignada	Máscara	1ª IP	última IP	Broadcast
LAN PC1	10.11.15.0	255.255.255.192	10.11.15.1	10.11.15.62	10.11.15.63
LAN PC2	10.11.15.64	" .224	" .65	" .94	" .95
LAN PC3	" " " .96	" .240	" .97	" .110	" .111
LAN PC4	" " .112	" .248	" .113	" .118	" .119
Enlace R1-R2	" " .120	" .252	" .121	" .122	" .123
Enlace R2-R3	" " .124	" .252	" .125	" .126	" .127

$\text{lan PC1} \Rightarrow \begin{cases} 2^n - 2 \geq 50 \\ 2^6 - 2 \geq 50 \\ 64 - 2 \geq 50 \\ 62 \geq 50 \end{cases} \left. \begin{array}{l} \text{máscara: } 32 - n \\ 32 - 6 \\ /26 \\ " " " .192 \end{array} \right\} \text{salto: } 2^n \rightarrow 2^6 = 64$

$\text{lan PC2} \Rightarrow \begin{cases} 2^n - 2 \geq 25 \\ 2^5 - 2 \geq 25 \\ 30 \geq 25 \end{cases} \left. \begin{array}{l} 32 - 5 \\ \text{mask: } /27 \\ .224 \end{array} \right\} \text{salto: } 2^n \rightarrow 2^5 = 32$

$\text{lan PC3} \Rightarrow \begin{cases} 2^n - 2 \geq 12 \\ 2^4 - 2 \geq 12 \\ 14 \geq 12 \end{cases} \left. \begin{array}{l} 32 - 4 \\ \text{mask: } /28 \\ .240 \end{array} \right\} \text{salto: } 2^n \rightarrow 2^4 = 16$

$\text{lan PC4} \Rightarrow \begin{cases} 2^n - 2 \geq 6 \\ 2^3 - 2 \geq 6 \\ 6 \geq 6 \end{cases} \left. \begin{array}{l} 32 - 3 \\ \text{mask: } /29 \\ .248 \end{array} \right\} \text{salto: } 2^n \rightarrow 2^3 = 8$

$\text{Enlace R1-R2} \Rightarrow \begin{cases} 2^n - 2 \geq 2 \\ 2^2 - 2 \geq 2 \\ 2 \geq 2 \end{cases} \left. \begin{array}{l} 32 - 2 \\ \text{mask: } /30 \\ .252 \end{array} \right\} \text{salto: } 2^n \rightarrow 2^2 = 4$

$\text{Enlace R2-R3} \Rightarrow 2 \geq 2 \quad .252$

PC1
PC2
PC3
R2

R2 < R1
R3

R3 < R2
PC4

Dispositivo	Interface	IP	Máscara	Gateway
R1	E0/0	10.11.15.1	255.255.255.192	N/A
R1	E0/1	" .65	" .224	N/A
R1	S1/0	" .97	" .248	N/A
R1	S1/0	" .121	" .252	N/A
R2	S1/0	" .122	" .252	N/A
R2	S1/1	" .125	" .252	N/A
R3	S1/1	" .126	" .252	N/A
R3	E0/0	" .113	" .248	N/A
PC1	NIC	" .2	" .192	10.11.15.1
PC2	NIC	" .66	" .224	" .65
PC3	NIC	" .98	" .240	" .97
PC4	NIC	" .114	" .248	" .113

- Trace
- Trace
- Show ip
- ip PC
- Show route

```

    10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S    10.11.15.0/26 [1/0] via 10.11.15.125
S    10.11.15.64/27 [1/0] via 10.11.15.125
S    10.11.15.96/28 [1/0] via 10.11.15.125
C    10.11.15.124/30 is directly connected, Serial1/1
S    10.11.15.120/30 [1/0] via 10.11.15.125
C    10.11.15.112/29 is directly connected, FastEthernet0/0

```

[R3_00111522#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Prot
FastEthernet0/0	10.11.15.113	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial1/0	unassigned	YES	unset	administratively down	down
Serial1/1	10.11.15.126	YES	manual	up	up
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down

R3_00111522#